

Juggernaut Router

Hybrid routing agent for AMD Developer Hackathon Track 1.

- Accuracy-first task routing across eight categories
- Deterministic answers when the pattern is safe
- Optional local GGUF inference
- Fireworks fallback through the injected base URL

The Problem

Sending every task to the same remote model is simple, but inefficient.

- Easy tasks waste tokens
- Hard tasks need stronger models
- Model output still needs validation
- The evaluator requires exact '/input' and '/output' behavior

The Solution

Juggernaut Router classifies each task, chooses a route, validates the answer, and writes evaluator-compatible JSON.

Route order:

1. Deterministic solver
2. Optional local model
3. Fireworks model fallback
4. Safe fallback only when no model route is available

Track 1 Coverage

The router supports all eight required categories.

- Factual knowledge
- Mathematical reasoning
- Sentiment classification
- Text summarisation
- Named entity recognition
- Code debugging
- Logical reasoning
- Code generation

Reliability

Accuracy comes from routing plus validation.

- Strict numeric and label normalization
- Code-shape validation
- Repetition and malformed-answer rejection
- Startup and finish logs for evaluator diagnostics
- Optional per-task telemetry for route analysis

Efficiency

The system optimizes tokens after correctness is protected.

- Avoid remote calls for locally safe tasks
- Use category-specific model and prompt policies
- Keep Fireworks as the quality path for risky tasks
- Keep Docker reproducible and linux/amd64 compatible

Demo Flow

1. Build the Docker image
2. Run synthetic sample tasks
3. Validate 'results.json'
4. Inspect route logs
5. Show the Cloudflare Worker demo page

Closing

Juggernaut Router is practical: classify the work, answer safe tasks cheaply, reserve stronger models for risky work, and validate everything before returning it.